

La genèse du mouvement Agile

Pourquoi et Comment naît le mouvement Agile ?

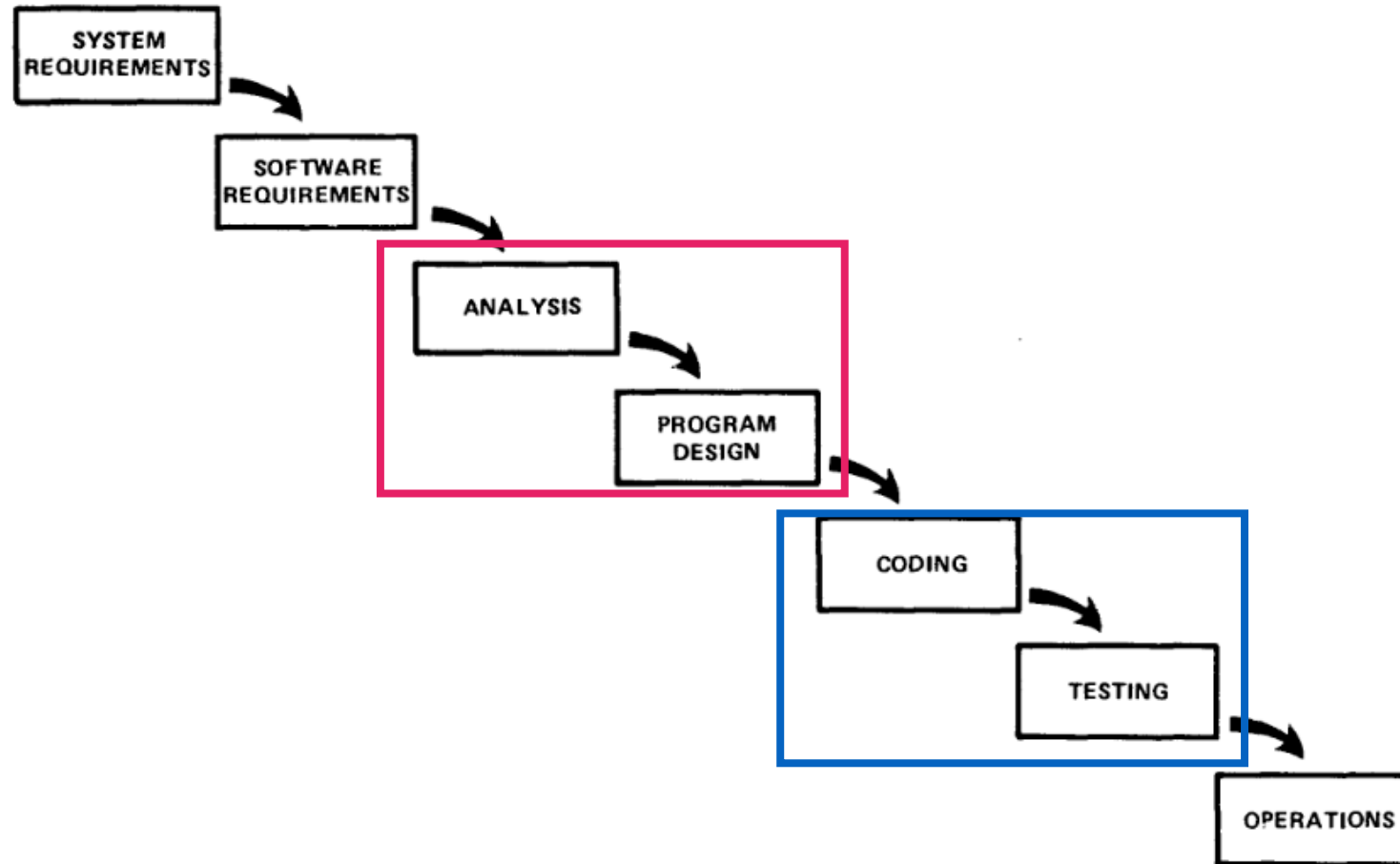
01 Avant Agile

01 Waterfall

Waterfall une méthode déjà itérative

Waterfall

Un système itératif dès les années 1970



une approche qui repousse
l'activité de production
jusqu'à ce qu'une phase
d'analyse est de planning
soit réalisée

Figure 2. Implementation steps to develop a large computer program for delivery to a customer.

*I believe in this concept, but the implementation described above is
risky and invites failure.*

MANAGING THE DEVELOPMENT OF LARGE SOFTWARE SYSTEMS

Dr. Winston W. Royce - 1970

Une phase de test trop tardive

Il met en évidence l'un des principaux problèmes du modèle en cascade : la phase de test intervient à la fin du processus de développement.

The testing phase which occurs at the end of the development cycle is the first event for which timing, storage, input/output transfers, etc., are experienced as distinguished from analyzed.

Implique de repenser le design

Ce qui peut conduire à des effets dévastateurs

The required design changes are likely to be so disruptive that the software requirements upon which the design is based and which provides the rationale for everything are violated. Either the requirements must be modified, or a substantial change in the design is required. In effect the development process has returned to the origin and one can expect up to a 100-percent overrun in schedule and/or costs.

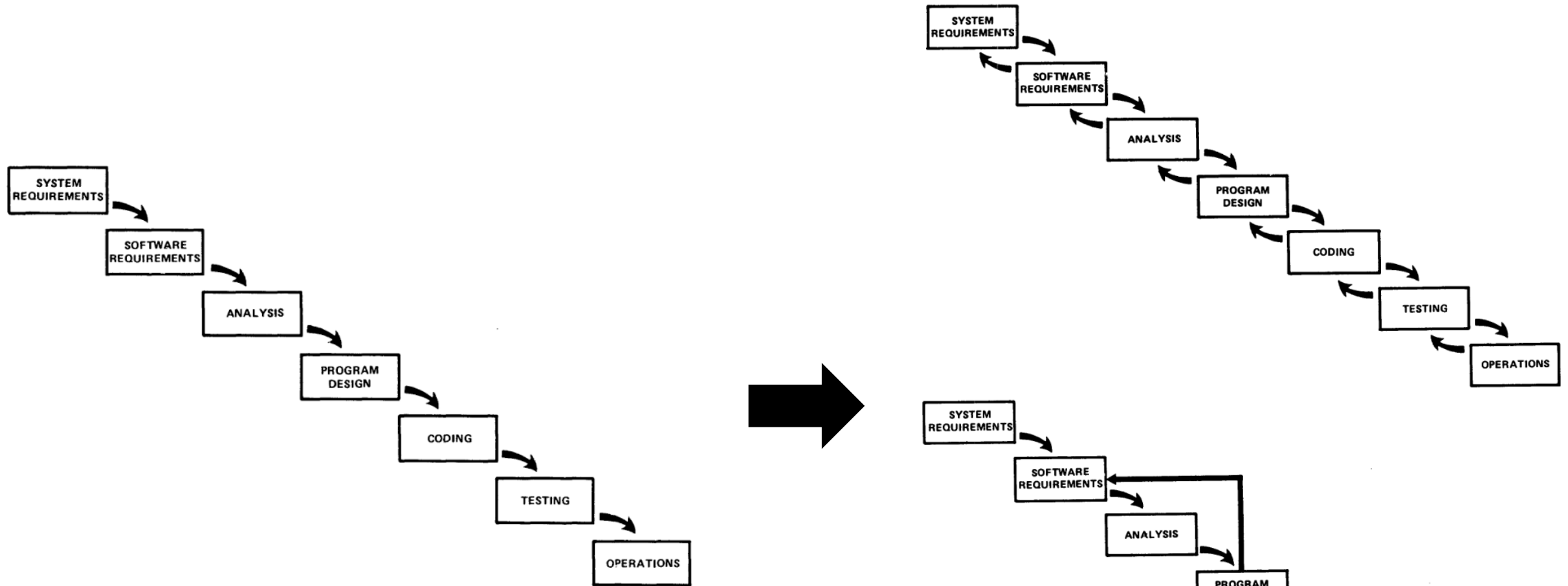


Figure 2. Implementation steps to develop a large computer program for delivery to a customer.

Nous avons donc des impacts forts entre le test → le code → l'analyse. **Les itérations ne sont donc pas cantonnées uniquement à la phase précédente.**

02 Waterfall est-il Agile ?

Pas totalement

- On retrouve dans le papier des concepts importants
 - Impliquer le client
 - Il critique le fait de tester à la fin

<http://www.bawiki.com/wiki/Waterfall.html>

<https://pragtob.wordpress.com/2012/03/02/why-waterfall-was-a-big-misunderstanding-from-the-beginning-reading-the-original-paper/>

03 Pourquoi Waterfall échoue

Pourquoi la mauvaise interprétation de waterfall échoue

Pourquoi s'être arrêté à la première interprétation ?

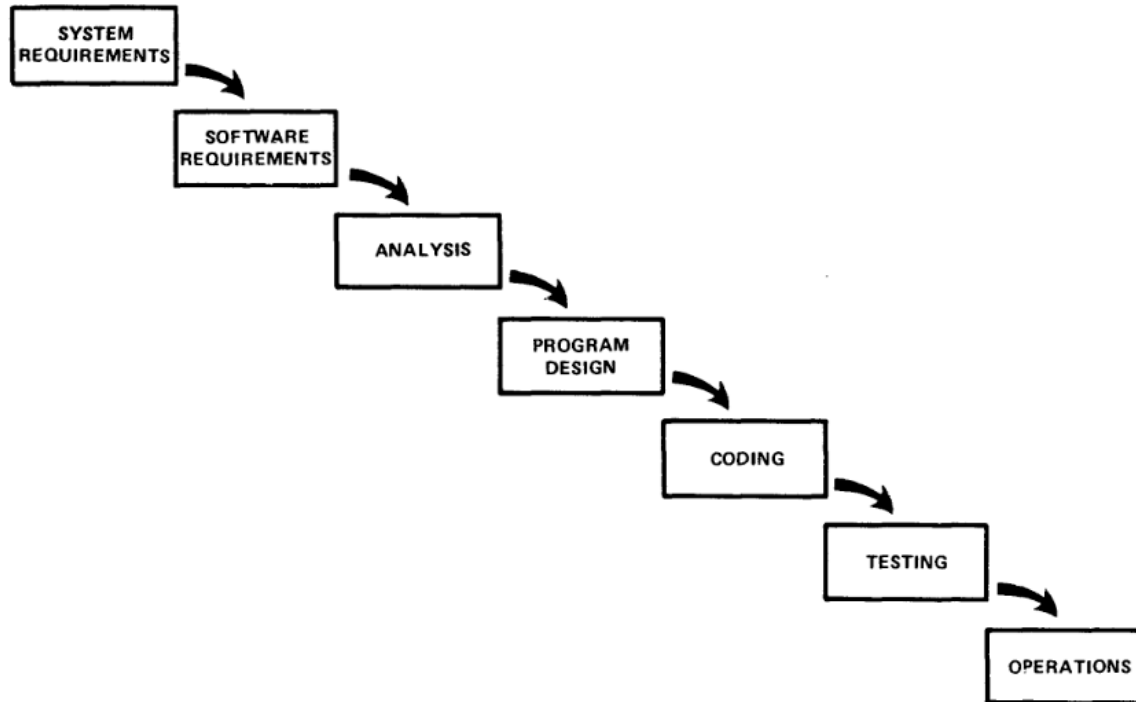
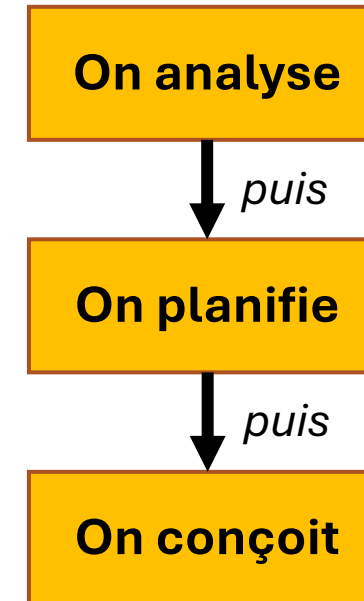


Figure 2. Implementation steps to develop a large computer program for delivery to a customer.

L'industrie logicielle va s'arrêter à cette figure car elle semble « NORMALE », « INTUITIVE » ou encore de « BON SENS »

En effet, faire les phases suivantes semble évident



The term "Waterfall" emerged later, likely due to misinterpretations and the influence of subsequent authors who cited Royce's work without fully understanding his criticisms.

Illusoire : Il n'existe qu'un ensemble fini de besoins

La phase d'analyse consiste à identifier **tous** les besoins

Même si nous passons une partie importante à pr
les besoins des utilisateurs il est très compliqué
exhaustif.

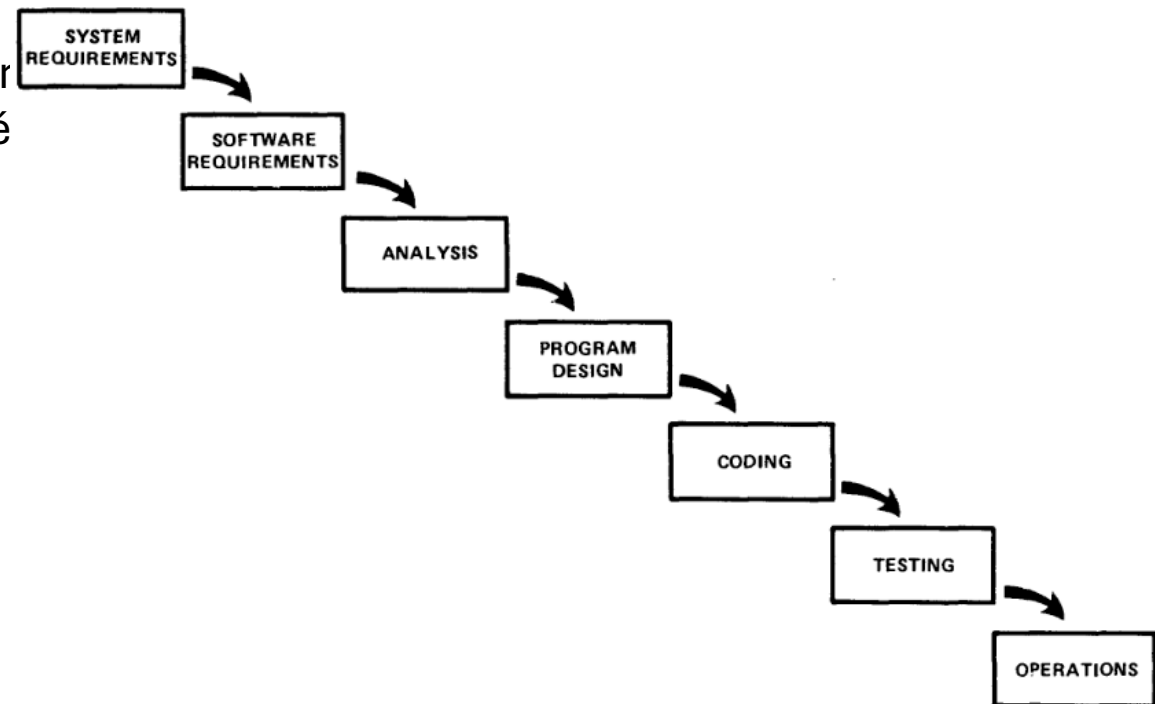


Figure 2. Implementation steps to develop a large computer program for delivery to a customer.

Illusoire : Les changements seront petits

Plus le délai entre la définition des besoins et la livraison du système est long, plus le changement sera important

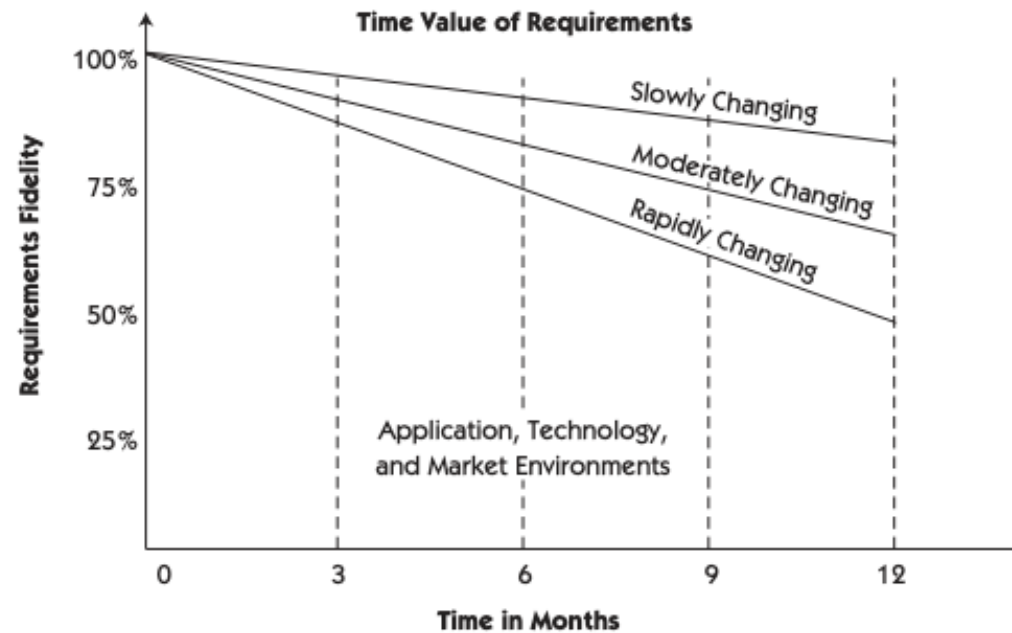


Figure 2-2 Time value of requirements

Illusoire : Délivrer dans les temps

- Nous avons planifié pour ce que nous savons et nous avons laissé suffisamment de place dans nos plans pour ce que nous ne savons pas.
- == à dire que la R&D et l'innovation peut être prédit ...

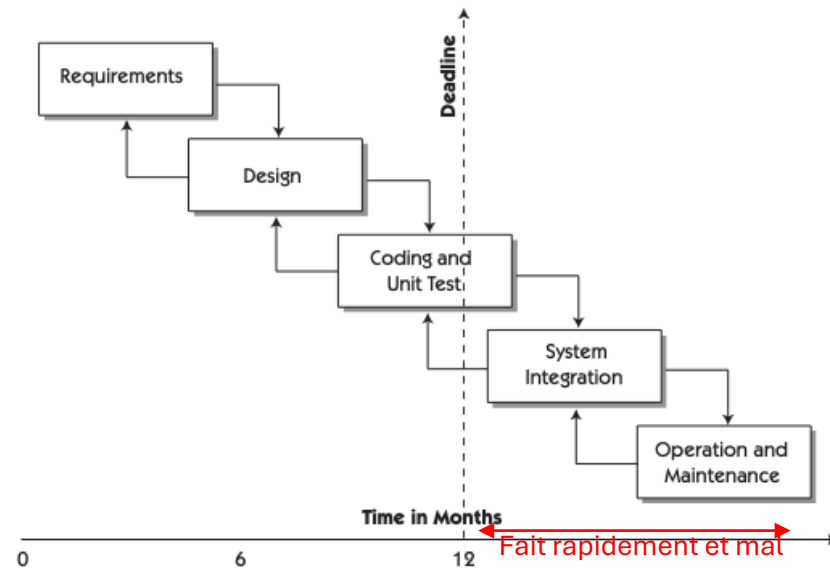


Figure 2-3 What happens when everything except the deadline moves to the right

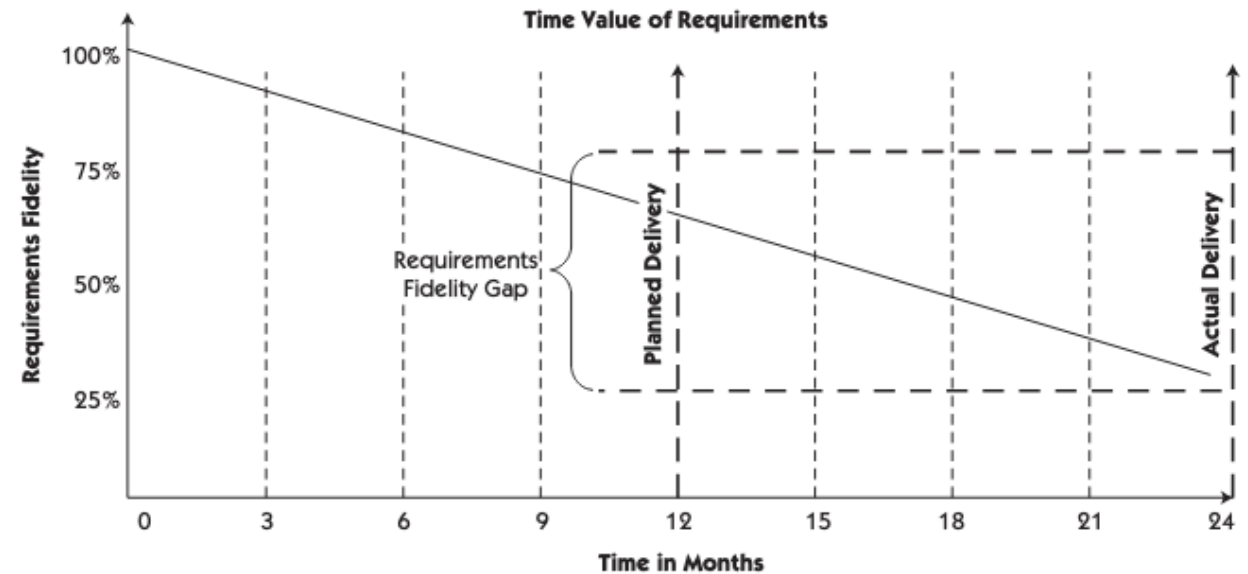


Figure 2-4 Planned versus actual requirements fidelity gap

Problème : Nous définissons plusieurs fois l'objectif du logiciel

- différentes équipes essayant de décrire complètement le produit logiciel cible trois fois: dans les spécifications détaillées, dans le code, et encore une fois dans les tests

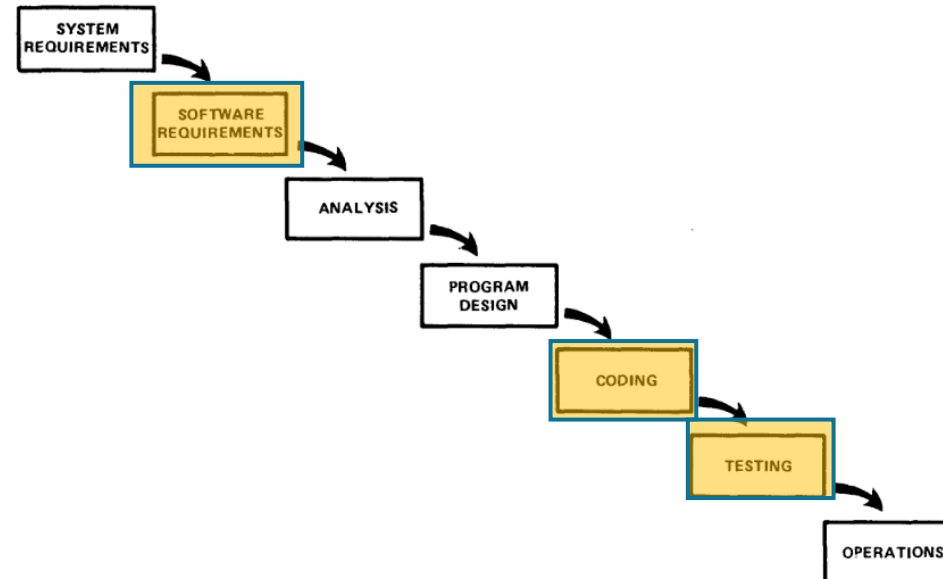


Figure 2. Implementation steps to develop a large computer program for delivery to a customer.

02 Agile

Le manifeste en 2001 mais une création dans les années 90'

Timeline des frameworks de gestion de projet

QUESTION

Pouvez-vous me donner les dates de Scrum, eXtreme Programming et Agile ?

1970

Waterfall

1986

The New New Product
Development Game
Hirotaka Takeuchi
Ikujiro Nonaka

1995

Scrum
Ken Schwaber

1999

eXtreme Programming
Kent Beck

2001

Manifeste Agile
17 signataires

01 Rencontre des 17 signataires

So Martin and I (Robert C. Martin) met in the fall of 2000 at a coffee shop near the ThoughtWorks office where he worked. **I described to him my idea to get all the competing lightweight process advocates together to form a manifesto of unity.** Martin made several recommendations for an invitation list, and we collaborated on writing the invitation. I sent the invitation letter later that day. The subject was “**Light Weight Process Summit**”.

Clean Agile

Robert C. Martin

Agile nait des mouvements XP, Scrum, Crystal, BDD, etc



- XP
- Scrum
- Behavior DD
- Crystal
- Pragmatic Programmers

02 Pourquoi le terme « Agile »

During 2000 a number of articles were written that referenced the category of “**Light**” or “**Lightweight**” processes. A number these articles referred to “Light methodologies, such as Extreme Programming, Adaptive Software Development, Crystal, and SCRUM”

Manifeste Agile 2001

<https://agilemanifesto.org/history.html>

Le terme Lightweight

The first edition of *Extreme Programming Explained: Embrace Change* had a definition of XP with the advantage of clarity: "XP is a lightweight methodology for small-to-medium-sized teams developing software in the face of vague or rapidly changing requirements." While this statement was true about the origin and intent of XP, it doesn't tell the whole story. In the five years since the publication of the first edition teams have pushed XP much further than the original definition. XP can be described this way:

- XP is lightweight. In XP you only do what you need to do to create value for the customer. You can't carry a lot of baggage and move fast. However, there is no freeze-dried software process. The body of technical knowledge necessary to be an outstanding team is large and growing.
- XP is a methodology based on addressing constraints in software development. It does not address project portfolio management, financial justification of projects, operations, marketing, or sales. XP has implications in all of these areas, but does not address these practices directly. Methodology is often interpreted to mean "a set of rules to follow that guarantee success." Methodologies don't work like programs. People

The name “Agile” was not a slam dunk. There were many different contenders. I happened to like “Light Weight,” but nobody else did. They thought it implied “inconsequential.” Others liked the word “Adaptive.” The word “Agile” was mentioned, and one person commented that it was currently a hot buzzword in the military. In the end, though nobody really loved the word “Agile,” it was just the best of a bunch of bad alternatives.

Clean Agile

Robert C. Martin

A retenir

- 1 La mauvaise interprétation de Waterfall a conduit à vouloir analyser complètement en amont

- 2 Agile naît des mouvements XP, Scrum etc où les auteurs constatent le besoin d'adaptation

- 3 Le terme *Lightweight* est d'abord utilisé pour définir les méthodologies qui s'adapteront rapidement aux changements des exigences.

Comprendre Agile c'est comprendre Scrum, XP, etc ...